Syntax-Aware Code Completion Extension for C in VSCode

Project Overview and Implementation

Background

- Visual Studio Code (VSCode) is a widely used IDE with a large extension ecosystem.
- Existing code completion often relies on IntelliSense and requires typing initial characters.
- Small Basic and C extensions lack deep syntactic and context-aware completion.

VSCode Extension for C

- Aimed at enhancing coding experience in C by integrating syntax-aware code completion.
- Supports context-based suggestions without requiring initial input.
- Integrates LR parsing with generative AI to suggest relevant syntax constructs.

Limitations of Existing Extensions

- Small Basic: Only syntax highlighting, lacks IntelliSense and debugging support.
- C: Requires typing multiple characters before any suggestions appear.
- Copilot: Relies on comments and prompts, may generate vague or incorrect suggestions.

How Our Extension Addresses These Limitations

- Provides syntax-aware code suggestions using LR parse states.
- Offers completions without initial user input.
- Displays top ranked syntax structures with explanation and structure preview.

Implementation Details

- Developed using VSCode Extension API with TypeScript and JavaScript.
- Uses YAPB-based LR parsers for Small Basic and C11 grammar.
- Prompts are constructed and sent to ChatGPT for suggestion generation.
- Suggestions are shown in a pop-up, and inserted into the editor upon selection.

Example 1: Code Completion in C

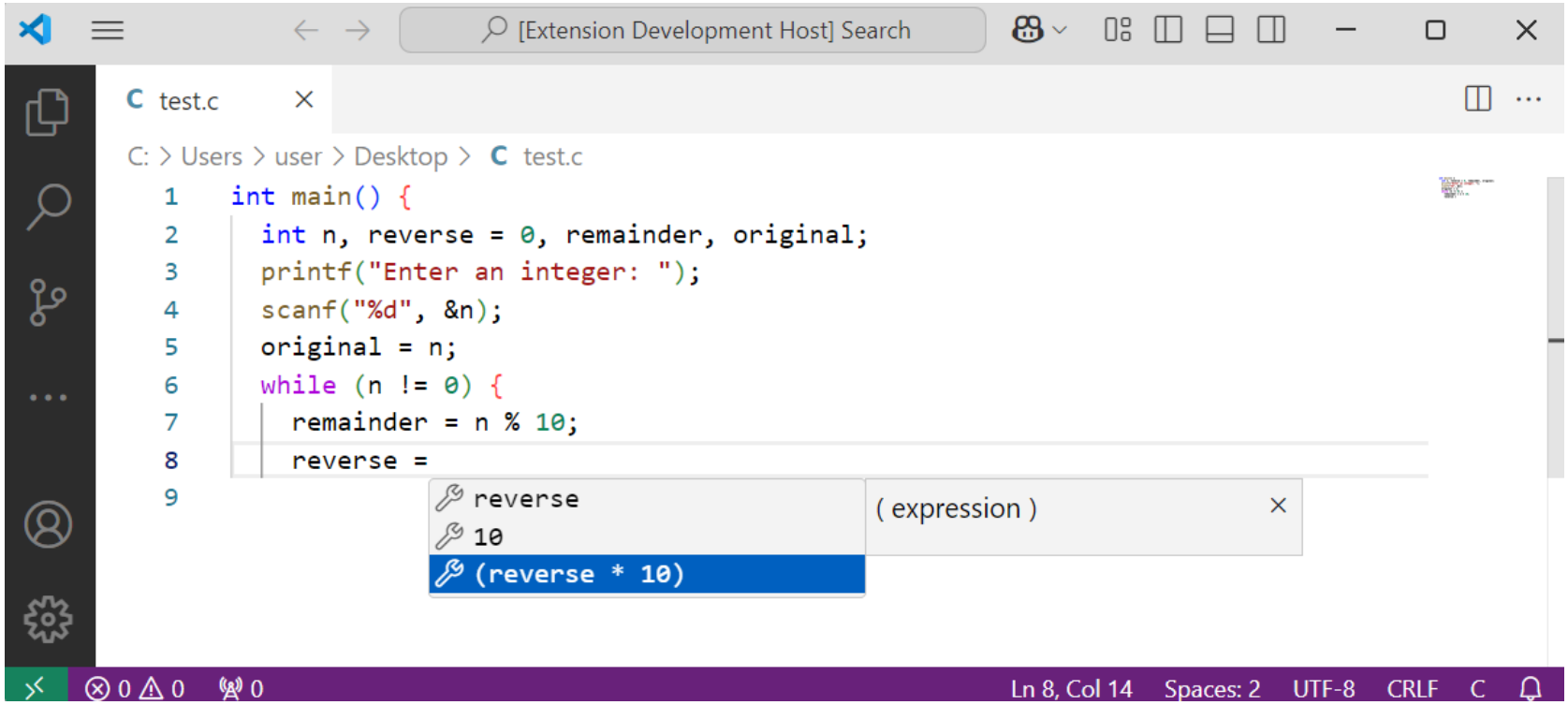


Figure: The VSCode extension suggesting '(celsius + step)' as a valid completion.

Example 2: Context-Aware Suggestions in C

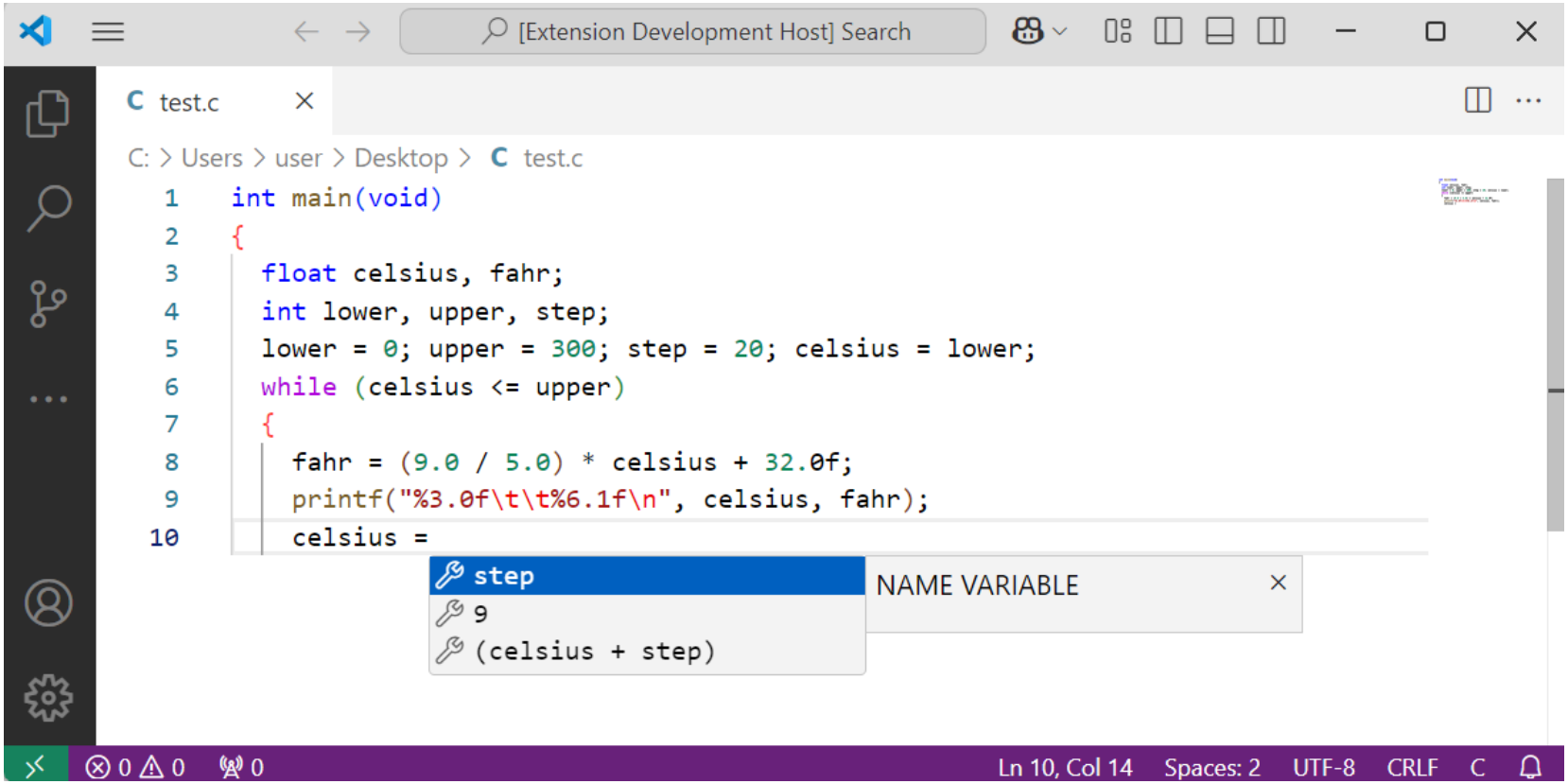


Figure: Syntax-based code completion showing '(reverse * 10)' in a loop context.

Thank You